11/8/25

Random Variables

Consider prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

Often, we will conduct an experiment & will be interested in some numerical property of the random outcome

Eg - Toss coin n times. $\Omega = \{\omega = (\omega_1, \omega_2 \ldots \omega_n), \omega_i \in \{H, T\}\}$.

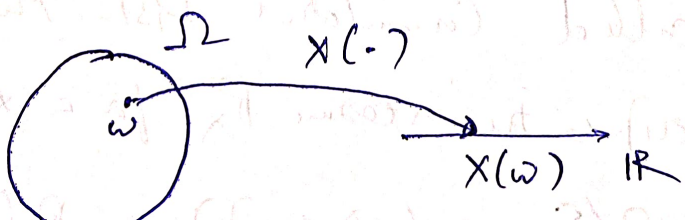
X = # heads observed $\quad X(HHH) = 3, \quad X(HTH) = 2$.

Eg Roll two dice $\quad \Omega = \{(i,j) : i,j \in [1,6]\}$.

X = sum of outcomes. $\quad X((1,6)) = 7$.

A random variable is essentially a fn. $X : \Omega \to \mathbb{R}$ (satisfying certain properties.

So in some sense, the term 'random variable' is a bit of a misnomer. It is a deterministic fn mapping outcomes from a prob. space to real numbers.



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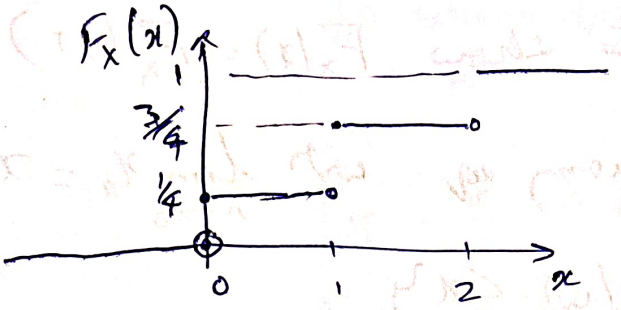
CDF

$(\Omega, \mathcal{F}, P)$, RV X , CDF $F_X(\cdot)$

Eg $\Omega = \{HH, HT, TH, TT\}$ $\quad X = \#$ heads.

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$

$1/4$ $1/4$ $1/4$ $1/4$



F_X monotone
starts at 0, goes to 1.
Can have jumps.

Thm - CDF F_X

(1) $x \le y \Rightarrow F_X(x) \le F_X(y)$

Pf $\{\omega : X(\omega) \le x\} \subseteq \{\omega : X(\omega) \le y\}$.

$\Rightarrow \mathbb{P}(\quad " \quad) \le \mathbb{P}(\quad " \quad)$

$F_X(x) \qquad F_X(y)$

(2) $\lim_{x \to -\infty} F_X(x) = 0$, $\lim_{x \to \infty} F_X(x) = 1$

Pf - suffices to show that for a strictly decreasing sequence $\{x_n\}$ s.t $\lim_{n\to\infty} x_n = -\infty$, $\lim_{n\to\infty} F_X(x_n) = 0$.

$$F_X(x_n) = \mathbb{P}(\underbrace{\{\omega : X(\omega) \le x_n\}}_{A_n})$$

A_n monotonically diminishing seq. of sets

$A_{m+1} \subseteq A_m$

$$\lim_{n\to\infty} F_X(x_n) = \lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcap_{n\ge 1} A_n\right) = 0$$

$\lim \mathbb{P}(A_n) = \mathbb{P}(\bigcap A_n)$: cont. of prob.

$\bigcap_{n\ge 1} A_n = \emptyset$

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(3) Similarly $\lim_{x \to -\infty} F_X(x) = 0$.

(4) $F_X(\cdot)$ is right continuous i.e. $\lim_{\varepsilon \downarrow 0} F_X(x+\varepsilon) = F_X(x)$ $\forall x$

Take $x \in \mathbb{R}$. Want to show $F_X(x) = F_X(x^+)$

Let $\{x_n\}$ be strictly decreasing seq with $\lim_{n \to \infty} x_n = x$.

$$A_n = \{\omega : X(\omega) \leq x_n\}$$

$$\mathbb{P}(A_n) = F_X(x_n)$$

As before, $A_{n+1} \subseteq A_n$ & $\bigcap_{n \geq 1} A_n = \{\omega : X(\omega) \leq x\}$

$$F_X(x) = \mathbb{P}\left(\bigcap_{n \geq 1} A_n\right) = \lim_{n \to \infty} \mathbb{P}(A_n) = \lim_{n \to \infty} F_X(x_n)$$

True for any such seq $\{x_n\}$.

CDF allows computation of prob. of interest

(1) $\mathbb{P}(X > x) = 1 - F_X(x)$

(2) $\mathbb{P}(X \in (a, b]) = F_X(b) - F_X(a)$

(3) $\mathbb{P}(X < x) = F_X(x^-)$

(4) $\mathbb{P}(X = x) = F_X(x) - F_X(x^-)$

$\mathbb{P}(X = x) > 0 \iff F_X$ is discontinuous at x & has jump of size $\mathbb{P}(X = x)$

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Types of RVs

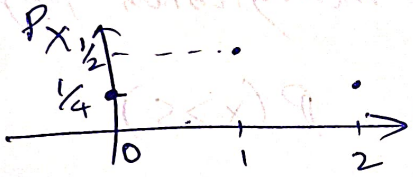
① Discrete RV – Support is a countable set
- ↳ finite
- ↳ Countably infinite

↙ Range of fn corresponding to RV.

PMF $P_X : \mathbb{R} \to [0,1]$

$P_X(x) = \mathbb{P}(X = x)$

For 2 coin tosses example, X = # heads

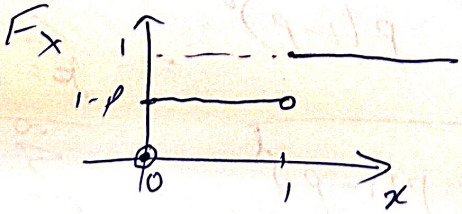


Zero everywhere else

$$F_X(x) = \mathbb{P}(X \le x) = \sum_{\substack{i \in S_X \\ i \le x}} P_X(i)$$

Eg ⓐ Bernoulli RVs – $\{0,1\}$

$P_X(1) = p, \quad P_X(0) = 1-p$



ⓑ Discrete Uniform RV $X \in \{a, a+1, \ldots, b\}$

$$P_X(i) = \frac{1}{b-a+1} \quad \forall i \in \{a, a+1, \ldots, b\}$$

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(c) **Binomial Dist** — parameterized by (n, p)

$X \in \{0, 1, \ldots, n\}$

$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$ for $k \in \{0, 1, \ldots, n\}$

(d) **Geometric dist** "no of coin tosses before first heads"

Support $= \mathbb{N}$

$P_X(k) = p(1-p)^{k-1}$

Geometric dist has the memorylessness property

Lowest sum heads in 100 tosses, prob it will take atleast 10 more vs $P(>10 \text{ tosses})$

$P(X > s+t \mid X > t) = P(X > s)$

Pf → $\dfrac{P(X > s+t \cap X > t)}{P(X > t)}$

$= \dfrac{P(X > s+t)}{P(X > t)}$

$= \dfrac{\sum_{j=s+t}^{\infty} p(1-p)^j}{\sum_{l=t}^{\infty} p(1-p)^l} \quad \xrightarrow{r = j-s} \quad \dfrac{\sum_{r=t}^{\infty} p(1-p)^{r+s}}{\sum_{l=t}^{\infty} p(1-p)^l}$

$= (1-p)^s$

RHS $= P(X > s) = \sum_{l=s}^{\infty} p(1-p)^l = (1-p)^s$

$= (1-p)^s \sum_{k=0}^{\infty} p(1-p)^k$

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→ Among discrete RVs, Geometric dist is the only one with memorylessness property. Applications in Queueing theory.

Poisson – Parameter $\lambda > 0$ $\quad X \in \mathbb{N}$

$$P_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k \in \mathbb{N}$$

(2) Continuous RVs

RV X is said to be cont^{n} if $\exists$ a non-negative fn $f_X : \mathbb{R} \to [0, \infty)$ s.t.

$$F_X(x) = \int_{-\infty}^{x} f_X(z)\, dz \qquad \forall x \in \mathbb{R}$$

$f_X(x)$ is called prob. density fn of X (pdf)

(a) F_X is Continuous. Also F_X is diff. almost everywhere & $\frac{d}{dx}F(x) = f_X(x)$

(b) $\int_{-a}^{\infty} f_X(x)\, dx = \lim_{x \to \infty} F_X(x) = 1$ [Integral of pdf gives prob. not $f_X(\cdot)$ itself]

(c) $P(a \leq X \leq b) = \int_a^b f_X(x)\, dx$

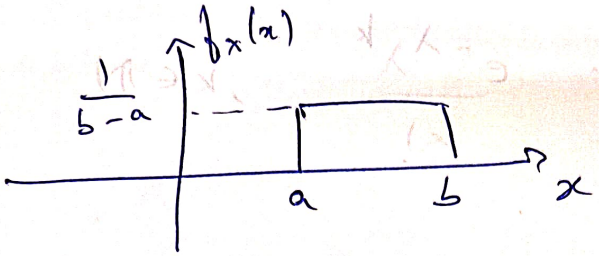
(d) $P(X = x) = 0 \qquad \forall x \in \mathbb{R}$.

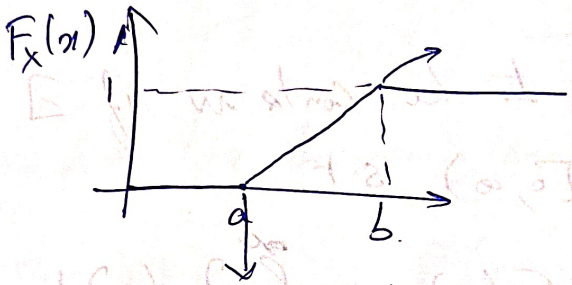
In fact, prob. that X assigns to any countable set is zero

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Eg ⓐ Uniform $[a, b]$

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{o.w} \end{cases}$$

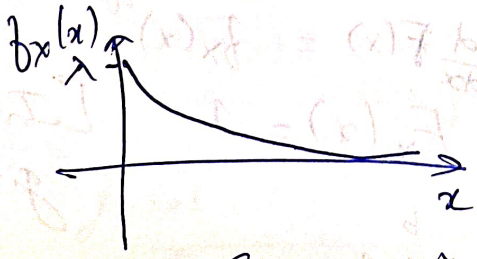


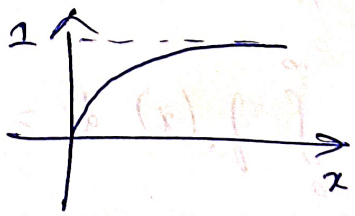


not diff. here.

ⓑ Exponential RV (λ) $\lambda > 0$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \ge 0 \\ 0 & \text{o.w} \end{cases}$$





$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x \ge 0 \\ 0 & x < 0 \end{cases}$$

Exp is only cont RV which is memoryless

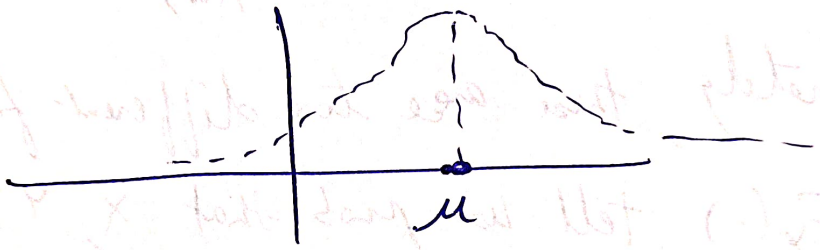
$$P(X > s+t \mid X > t) = P(X > s)$$ [Prove it]

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(3) **Gaussian RV / Normal RV**

(μ, σ^2) $\mu \in \mathbb{R}$, $\sigma > 0$

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$



$\mu = 0$
$\sigma = 1$
↓
standard Normal dist

CDF doesn't have closed form expression computed numerically.

(4) Above are examples of light tailed distn.
Heavy tailed dist. have significant mass away from mean

Eg Cauchy dist $\quad f_X(x) = \frac{1}{\pi(1+x^2)} \quad x \in \mathbb{R}$

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